

CA5369: Use XmlReader for Deserialize (code analysis)

CA5369: Use XmlReader for Deserialize (code analysis) - filipsebesta, learn.microsoft.com.

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Content

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CA5369: Use XmlReader for Deserialize

Summarize this article for me

Property Value	Rule ID CA5369	Title Use XmlReader for Deserialize
Category Security	Fix is breaking or non-breaking Non-breaking	Enabled by default in .NET 10 No
Applicable languages C# and Visual Basic		

Cause

Deserializing untrusted XML input with `XmlSerializer.Deserialize` instantiated without an `XmlReader` object can potentially lead to denial of service, information disclosure, and server-side request forgery attacks. These attacks are enabled by untrusted DTD and XML schema processing, which allows for the inclusion of XML bombs and malicious external entities in the XML. Only with `XmlReader` is it possible to disable DTD. Inline XML schema processing as `XmlReader` has the `ProhibitDtd` and `ProcessInlineSchema` property set to `false` by default in .NET Framework version 4.0 and later. The other options such as `Stream`, `TextReader`, and `XmlSerializationReader` cannot disable DTD processing.

Rule description

Processing untrusted DTD and XML schemas may enable loading dangerous external references, which should be restricted by using an `XmlReader` with a secure resolver or with DTD and XML inline schema processing disabled. This rule detects code that uses the [XmlSerializer.Deserialize](#) method and does not take `XmlReader` as a constructor parameter.

How to fix violations

Do not use [XmlSerializer.Deserialize](#) overloads other than [Deserialize\(XmlReader\)](#), [Deserialize\(XmlReader, String\)](#), [Deserialize\(XmlReader, XmlDeserializationEvents\)](#), or [Deserialize\(XmlReader, String, XmlDeserializationEvents\)](#).

When to suppress warnings

You can potentially suppress this warning if the parsed XML comes from a trusted source and hence cannot be tampered with.

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA5369
// The code that's violating the rule is on this line.
#pragma warning restore CA5369
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA5369.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Pseudo-code examples

Violation

The following pseudo-code sample illustrates the pattern detected by this rule. The type of the first parameter of [XmlSerializer.Deserialize](#) is not `XmlReader` or a derived class thereof.

```
using System.IO;
using System.Xml.Serialization;
...
new XmlSerializer(typeof(TestClass)).Deserialize(new FileStream("filename",
    FileMode.Open));
```

Solution

```
using System.IO;
using System.Xml;
using System.Xml.Serialization;
...
new XmlSerializer(typeof(TestClass)).Deserialize(XmlReader.Create (new
    FileStream("filename", FileMode.Open)));
```

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